FLOQUET ISOSPECTRALITY TO THE DISCRETE LAPLACIAN: LEVEL-SET DEGENERATIONS, PERIOD LATTICES AND NODE COORDINATES

KEIO

Abstract. Let Δ be the discrete Laplacian on $\mathbb{Z}^d$ and let $\Gamma \subset \mathbb{Z}^d$ be a full-rank period lattice. Faust, Friedman, O'Malley, Ramos and Sharma asked for the classification of the periods Γ admitting a nonzero complex Γ-periodic potential V such that $\Delta + V$ is Floquet isospectral to Δ. We introduce a *level-set degeneration* method: the Floquet multipliers are sent to infinity along complex directions chosen so that the leading diagonal of the Floquet matrix is constant on a prescribed level set K of the dual group. In the limit the potential acts on K by a group convolution, or by a fibred family of lower-dimensional Floquet problems, whose rigidity can be read off directly.

The method gives a short, computer-free proof of the rectangular classification (nonzero potentials exist if and only if some period exceeds 3); the rigid half of this classification was also obtained, by a perturbative pole-separation argument, in an unrefereed manuscript of the author. For arbitrary lattices we prove: nonrigidity whenever some axis index $[\mathbb{Z} : \pi_j(\Gamma)]$ is at least 4; rigidity of $\Gamma' \oplus 2\mathbb{Z}$ whenever Γ' is rigid; a recursive rigidity criterion, which gives for instance rigidity of $\Gamma' \oplus 3\mathbb{Z}$ for all diagonal lattices Γ'; and a reduction of the full classification to a single *off-axis vanishing* statement, together with exact identities for the minimal cycles of the Fourier support. We verify the expected classification by exact symbolic computation for all planar lattices of index at most 10 and for all cyclic lattices in $\mathbb{Z}^3$ of index at most 7.

For the one-dimensional Conjecture 2 of the same authors (existence of conjugate-reflection symmetric potentials Floquet isospectral to 0) we construct potentials from Baker–Akhiezer functions with prescribed data at the double points of the free spectral curve. For an odd period q this gives a sufficient condition for the conjecture: a nonzero real solution of an explicit square system of $q-1$ multilinear equations with positive coefficients of Hirota type, subject to one open condition. With interval Newton (Krawczyk) certificates in ball arithmetic for every prime q with $5 \le q \le 199$ and for $q = 9$, we obtain Conjecture 2 for every period that is even, divisible by 9, or divisible by a prime p with $5 \le p \le 199$; in particular for all periods $3 < q \le 210$. Neither the classification for general nonrectangular lattices nor Conjecture 2 for all odd periods is settled here; we state the expected answer (rigidity if and only if all axis indices are at most 3) and a stronger separation conjecture.

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1. Introduction

1.1. **The problem.** For $n \in \mathbb{Z}^d$ let $(\Delta u)(n) = \sum_{j=1}^{d} (u(n+e_j) + u(n-e_j))$. Let $\Gamma \subset \mathbb{Z}^d$ be a sublattice of finite index $N = [\mathbb{Z}^d : \Gamma]$ and let $V \colon \mathbb{Z}^d \to \mathbb{C}$ be Γ-periodic. For a character $\chi \colon \Gamma \to \mathbb{T}$ the operator $\Delta + V$ preserves the N-dimensional space of χ-quasiperiodic functions; its spectrum there is the Floquet spectrum $\sigma_V(\chi)$. The operators $\Delta + V$ and Δ are *Floquet isospectral* if $\sigma_V(\chi) = \sigma_0(\chi)$, with multiplicity, for every χ. We call Γ *rigid* if $V = 0$ is the only Γ-periodic complex potential Floquet isospectral to 0, and *nonrigid* otherwise.

For real potentials rigidity always holds (Kappeler [6]). For complex potentials the situation is different: Faust, Liu, Matos, Plute, Robinson, Tao, Tran and Zhuang [5] produced explicit nonzero examples, and Cobb, Faust and Kretschmer [3, Theorem C] proved that a rectangular period $\Gamma = q_1\mathbb{Z} \oplus \cdots \oplus q_d\mathbb{Z}$ is nonrigid as soon as some $q_i > 3$, and rigid when all $q_i \le 3$ and at most two of them equal 3. They identified $(3\mathbb{Z})^3$ as the smallest open case [3, Section 4.4]. Faust, Friedman, O'Malley, Ramos and Sharma [4] posed the general question:

Problem 1.1 ([4, Problem 1])**.** *For the $\mathbb{Z}^d$-periodic grid graph, classify all periods Γ for which there exist nonzero Γ-periodic potentials that are Floquet isospectral to the discrete Laplacian.*

They note that the problem is solved for $d = 1$, open for $d \ge 2$, and that even for rectangular periods the barrier is the case where all periods equal 3. For $d = 2$ they observe that the lattices not contained in $q\mathbb{Z}\oplus\mathbb{Z}$ or $\mathbb{Z}\oplus q\mathbb{Z}$ for some $q > 3$ are the Hermite normal forms $\Gamma = (a,0)\mathbb{Z}\oplus(b,c)\mathbb{Z}$ with $\gcd(a,b), c \le 3$ and report a symbolic check of all such lattices with $ac \le 8$; in every dimension they prove rigidity of the lattice generated by $(a,0,\dots,0)$, $(1,1,0,\dots,0), \dots, (1,0,\dots,0,1)$ [4, Theorem 6.1], which in $d = 2$ is $(a,0)\mathbb{Z} \oplus (1,1)\mathbb{Z}$.

In an earlier, unrefereed manuscript [8] (Part 12) the author gave an argument that every rectangular period with all $q_j \le 3$ is rigid, which together with [3, Theorem C] would settle the rectangular case of Problem 1.1. That argument combined coordinate averaging with quadratic and cubic Rayleigh–Schrödinger identities and a separation of their linear poles. The present paper develops a different, non-perturbative mechanism and applies it to arbitrary period lattices. The

nonrectangular case of Problem 1.1 is *not* solved here; we state precisely what is proved and what remains (Section 8).

1.2. **Notation.** Write $\theta^n = \prod_j \theta_j^{n_j}$ for $\theta \in (\mathbb{C}^*)^d$ and $n \in \mathbb{Z}^d$. The *dual group* of Γ is

$$H = H(\Gamma) = \{\theta \in (\mathbb{C}^*)^d : \ \theta^\gamma = 1 \text{ for all } \gamma \in \Gamma\},$$

a finite subgroup of $\mathbb{T}^d$ of order N, Pontryagin dual to $\mathbb{Z}^d/\Gamma$. Every Γ-periodic V has a unique expansion $V(n) = \sum_{h\in H} \widehat{V}(h)h^n$. For $j \in [d] = \{1, \dots, d\}$ let

$$H_j = \{\theta \in H : \theta_j = 1\}, \qquad A_j = \{\theta \in H : \theta_k = 1 \ (k \neq j)\}, \qquad H^\circ = H \setminus \bigcup_j H_j.$$

The *axis index* is $g_j(\Gamma) = |A_j| = [\mathbb{Z} : \pi_j(\Gamma)]$, where π_j is the j-th coordinate projection, and the *directional period* is $m_j(\Gamma) = |\pi_j(H)| = \min\{m \geq 1 : me_j \in \Gamma\}$; always $g_j \mid m_j$. For rectangular Γ one has $g_j = m_j = q_j$. We write $\pi_{\hat{\jmath}} \colon \mathbb{Z}^d \to \mathbb{Z}^{d-1}$ for the projection forgetting the j-th coordinate.

1.3. **Main results.**

Theorem A (Level-set degeneration; Theorem 3.1 and Lemma 3.7)**.** *Let V be Floquet isospectral to 0. Let $J \subset [d]$, $\sigma \in \{\pm 1\}^J$, $c \in (\mathbb{C}^*)^J$, and let $K \subset H$ be a level set of $\ell(\theta) = \sum_{j\in J} c_j\theta_j^{\sigma_j}$. Then for all $s \in \mathbb{C}^J$ and all $\xi \in (\mathbb{C}^*)^{[d]\setminus J}$ the $K \times K$ matrix $\operatorname{diag}\big(\sum_{j\in J} s_j\theta_j^{\sigma_j} + \sum_{j\notin J}(\xi_j\theta_j + \xi_j^{-1}\theta_j^{-1})\big)_{\theta\in K} + \big(\widehat{V}(\theta'\theta^{-1})\big)_{\theta',\theta\in K}$ has the same characteristic polynomial as its diagonal part. If K is a subgroup and $s = 0$, this splits into a family of lower-dimensional Floquet isospectrality problems indexed by the characters of $K \cap \{\theta_j = 1,\ j \notin J\}$.*

Choosing $J = \{j\}$ recovers the known fact that directional averages of an isospectral potential are isospectral (Corollary 3.3); choosing $J = [d]$ with a "balanced" direction makes K a cyclic group on which $\widehat{V}$ must vanish (Corollary 3.6). This alone gives a one-paragraph proof that $(3\mathbb{Z})^d$ is rigid for all d, and, with the fibration part of Theorem A, a computer-free proof of:

Theorem B (Rectangular periods; Theorems 4.1 and 4.2)**.** *$q_1\mathbb{Z} \oplus \cdots \oplus q_d\mathbb{Z}$ is rigid if and only if $\max_j q_j \leq 3$.*

The nonrigid direction is [3, Theorem C], and the rigid direction was also obtained, by a different and perturbative argument, in the unrefereed manuscript [8]. The proof given here uses no perturbation expansion and no symbolic computation, and it extends beyond rectangular periods:

Theorem C (Arbitrary periods; Section 5)**.** *Let $\Gamma \subset \mathbb{Z}^d$ be any full-rank lattice.*

(i) *If $g_j(\Gamma) \geq 4$ for some j, then Γ is nonrigid.*
(ii) *For every lattice $\Gamma' \subset \mathbb{Z}^{d-1}$, $\Gamma' \oplus 2\mathbb{Z}$ is rigid if and only if Γ' is rigid.*
(iii) *If $m_j(\Gamma) \leq 3$ for all j, then Γ is rigid.*
(iv) (Criterion) *If all projections $\pi_j(\Gamma)$ are rigid and every $\eta \in H^\circ$, with the possible exception of one pair $\{h, h^{-1}\}$, lies in a "balanced" subgroup whose fibred lattice is rigid (Definition 5.4), then Γ is rigid. This recovers the rigidity of the diagonal lattices $\Gamma_{\varepsilon,N} = \{n : \sum_j \varepsilon_j n_j \equiv 0 \bmod N\}$ [4, Theorem 6.1] and shows that $\Gamma_{\varepsilon,N} \oplus 3\mathbb{Z}$ is rigid for all ε and N (Example 5.7); the directional periods of these lattices are unbounded.*

Part (i) is a direct extension of an observation made in [4, Section 6] for $d = 2$, and part (iii) is an immediate consequence of Theorem B; parts (ii) and (iv) appear to be new for nonrectangular lattices. Theorem C(i) says that the condition $\max_j g_j(\Gamma) \leq 3$ is necessary for rigidity. We expect it to be sufficient:

Conjecture 1.2 (Expected answer to Problem 1.1)**.** *A full-rank lattice $\Gamma \subset \mathbb{Z}^d$ is rigid if and only if $g_j(\Gamma) = [\mathbb{Z} : \pi_j(\Gamma)] \leq 3$ for every j.*

Theorem D (Reduction; Theorem 5.9, Remark 5.10 and Proposition 5.11)**.** *Conjecture 1.2 holds in all dimensions if and only if, for every $d \geq 2$ and every $\Gamma \subset \mathbb{Z}^d$ with $\max_j g_j(\Gamma) \leq 3$, every Floquet isospectral potential with $\operatorname{supp} \widehat{V} \subset H^\circ$ vanishes. For such potentials, $\widehat{V}(h)\widehat{V}(h^{-1}) = 0$ for every $h \in H^\circ$, and in dimension 2 the minimal-length cycles of the support satisfy explicit residue identities.*

Theorem E (Computations; Section 7)**.**
(a) *Every planar lattice $(a,0)\mathbb{Z} \oplus (b,c)\mathbb{Z}$, $0 \leq b < a$, with $2 \leq ac \leq 10$ and $\gcd(a,b), c \leq 3$ is rigid* (64 lattices; exact computation over cyclotomic fields).
(b) *Every lattice in $\mathbb{Z}^3$ with cyclic dual group of order at most 7 and all axis indices at most 3 is rigid.*
(c) *For $\Gamma = 4\mathbb{Z} \oplus 2\mathbb{Z}$ every Floquet isospectral potential is a function of n_1 alone.*
(d) *For every prime q with $5 \leq q \leq 199$ and for $q = 9$ there is a nonzero $q\mathbb{Z}$-periodic potential $(v_1, v_2, \ldots, \overline{v_2}, \overline{v_1})$ Floquet isospectral to 0* (Krawczyk certificates in ball arithmetic)*. Consequently Conjecture 2 of [4] holds for every $q > 3$ that is even, divisible by 9, or divisible by a prime p with $5 \leq p \leq 199$; in particular for all $3 < q \leq 210$.*

For one-dimensional potentials we also give an explicit construction. Let $q = 2m+1$ be odd, $\theta_l = e^{\pi i l/q}$ for $1 \leq l \leq 2m$, and $\omega = e^{2\pi i/q}$; the points $\theta_l^{\pm 1}$ are the preimages of the double points of the free spectral curve. Put $a_{lk} = \sin^2(\pi(k-l)/2q)/\sin^2(\pi(k+l)/2q)$, $A_I = \prod_{l<k\in I} a_{lk}$ and $s(I) = \sum_{l\in I} l$ for $I \subset \{1, \ldots, 2m\}$.

Theorem F (Node coordinates; Theorem 6.2 and Corollary 6.3)**.** *If $\beta \in \mathbb{C}^{2m}$ is such that the polynomial $\tau(z) = \sum_I A_I \beta^I z^{s(I)}$ agrees on the q-th roots of unity with Cz^{ν_0} for some $C \neq 0$ and $\nu_0 \in \mathbb{Z}/q$, then the Baker–Akhiezer function with node data β defines a q-periodic potential V, given by an explicit ratio of such sums, which is Floquet isospectral to 0, is nonzero when $\beta \neq 0$, and satisfies $V_{-1-n} = \overline{V_n}$ when β is real. Hence, for an odd period q, a sufficient condition for Conjecture 2 is a nonzero real solution β of the square system of $q-1$ multilinear equations with positive coefficients expressing that τ has no Fourier mode on $\mathbb{Z}/q$ other than ν_0, together with the open condition that the ν_0-mode does not vanish.*

The construction is the discrete analogue of the description of finite-gap potentials by Baker–Akhiezer functions on a singular spectral curve; the point is that for the free curve all node data are explicit and the Schrödinger condition (constant off-diagonal entries) becomes the single statement that τ has one Fourier mode on $\mathbb{Z}/q$. All real solutions we found numerically arise in this way (Remark 6.5).

1.4. **Relation to earlier work.** Asymptotic analysis of Bloch and Fermi varieties at infinity is a central tool in the work of Liu [9, 10, 11], and coordinatewise averaging (our Corollary 3.3) appears explicitly in [8, Part 12]. The new ingredient here is to degenerate along complex directions that are *resonant* on a prescribed level set of the dual group, together with the observation that a resonant level set which is a subgroup turns the potential into a group convolution or a fibred family of Floquet problems. The real case of separability is due to Liu [11]; complete Floquet invariants for periodic graphs are due to Saburova [12]; the algebraic background is surveyed by Shipman and Sottile [16]. Brysiewicz, Faust and Liu [2] recently showed that Fermi isospectral rigidity fails in dimension two. The author's note [7] excludes nonzero $(3\mathbb{Z})^3$-periodic shifts of support at most six at the periodic fibre; that partial result is superseded by Theorem B.

Besides Problem 1.1, [4] states two existence conjectures in dimension one: nonzero isospectral potentials of the reflection-antisymmetric form $(v_1, v_2, \ldots, -v_2, -v_1)$ (Conjecture 1) and of the conjugate-reflection form $(v_1, v_2, \ldots, \overline{v_2}, \overline{v_1})$ (Conjecture 2), both proved there for even periods (Conjecture 1 when $4 \mid q$). An argument for Conjecture 1 for all periods, comparing the multiplicity of the origin with a Bezout number on the antisymmetric slice, is given in the author's unrefereed

manuscript [8, Part 13]. Conjecture 2 asks for *real* points of a real form, which counting arguments of that kind do not detect, and it is the one-dimensional question addressed in Section 6. Baker–Akhiezer functions for periodic difference operators go back to van Moerbeke and Mumford [15]; our use of them is elementary and self-contained.

1.5. **Organization.** Section 2 sets up the Floquet normal form for an arbitrary period lattice. Section 3 proves Theorem A and its corollaries. Section 4 proves Theorem B. Section 5 treats arbitrary lattices (Theorems C and D). Section 6 proves Theorem F. Section 7 describes the computations (Theorem E), and Section 8 lists what remains open.

2. Floquet normal form for an arbitrary period

For $w \in (\mathbb{C}^*)^d$ put $\lambda(w) = \sum_{j=1}^{d}(w_j + w_j^{-1})$.

Lemma 2.1 (Twisted Fourier basis)**.** *Let $\zeta \in (\mathbb{C}^*)^d$ and let $\chi_\zeta(\gamma) = \zeta^\gamma$. The functions $e_\theta(n) = (\zeta\theta)^n$, $\theta \in H$, form a basis of the space of χ_ζ-quasiperiodic functions, and*

$$(\Delta + V)e_\theta = \lambda(\zeta\theta)\, e_\theta + \sum_{\theta' \in H} \widehat{V}(\theta'\theta^{-1})\, e_{\theta'}. \tag{1}$$

Hence the Floquet matrix in this basis is $M_V(\zeta) = D(\zeta) + C_V$, where $D(\zeta) = \operatorname{diag}(\lambda(\zeta\theta))_{\theta\in H}$ and $(C_V)_{\theta'\theta} = \widehat{V}(\theta'\theta^{-1})$.

Proof. $e_\theta(n+\gamma) = \zeta^\gamma\theta^\gamma e_\theta(n) = \chi_\zeta(\gamma)e_\theta(n)$. The e_θ are independent because the characters $n \mapsto \theta^n$ of $\mathbb{Z}^d/\Gamma$ are distinct, and there are N of them. The formula follows from $\Delta(\zeta\theta)^n = \lambda(\zeta\theta)(\zeta\theta)^n$ and $V(n)e_\theta(n) = \sum_h \widehat{V}(h)e_{h\theta}(n)$. □

The map $\zeta \mapsto \chi_\zeta$ from $(\mathbb{C}^*)^d$ onto $\operatorname{Hom}(\Gamma, \mathbb{C}^*)$ is surjective with fibres the cosets ζH, and $\zeta \in \mathbb{T}^d$ gives exactly the unitary characters. Put $P_V(x, \zeta) = \det(x - M_V(\zeta))$.

Lemma 2.2. *V is Floquet isospectral to 0 if and only if $P_V(x, \zeta) = P_0(x, \zeta)$ as polynomials in x with Laurent polynomial coefficients in ζ. In that case the identity holds for every $\zeta \in (\mathbb{C}^*)^d$.*

Proof. Isospectrality is the identity for $\zeta \in \mathbb{T}^d$, and a Laurent polynomial vanishing on $\mathbb{T}^d$ is zero. □

We record the symmetries used below. For $g \in H$ let $T_g e_\theta = e_{g\theta}$. Then $M_V(\zeta g) = T_g^{-1} M_V(\zeta) T_g$, so $P_V(x, \zeta g) = P_V(x, \zeta)$. Since $\operatorname{tr} M_V = \operatorname{tr} D + N\widehat{V}(1)$, every isospectral potential satisfies

$$\widehat{V}(1) = 0. \tag{2}$$

Lemma 2.3 (Sublattices)**.** *If $\Gamma' \subset \Gamma$ has finite index and V is Γ-periodic and Floquet isospectral to 0 with respect to Γ, then it is so with respect to Γ'.*

Proof. $H(\Gamma') \supset H(\Gamma)$ and, as a Γ'-periodic function, V has Fourier support in $H(\Gamma)$. Hence $M_V^{\Gamma'}(\zeta)$ is block diagonal along the cosets $\theta_i H(\Gamma)$, with blocks $M_V^{\Gamma}(\zeta\theta_i)$. □

Lemma 2.4 (Irreducibility of the free Floquet polynomial)**.** *Let $k \geq 0$, let $G \subset \mathbb{T}^k$ be a finite subgroup with $\Lambda = G^\perp \subset \mathbb{Z}^k$, and let $z = (\zeta^{\mu_1}, \dots, \zeta^{\mu_k})$ for a basis (μ_i) of Λ. Then $F_G(x, \zeta) = \prod_{g\in G}(x - \lambda(\zeta g))$ lies in $\mathbb{C}[z^{\pm 1}][x]$ and is irreducible there. The same holds for the characteristic polynomial of any Floquet matrix $M_W^G(\zeta)$ of a Λ-periodic potential W, except irreducibility. For $k = 0$ we read $F_G = x$.*

Proof. The group G acts on $\mathbb{C}(\zeta)$ by $\zeta \mapsto \zeta g$; a monomial ζ^n is invariant iff $n \in \Lambda$, so $\mathbb{C}[\zeta^{\pm 1}]^G = \mathbb{C}[z^{\pm 1}]$ and $\mathbb{C}(\zeta)^G \supset \mathbb{C}(z)$. Choose a basis $f_1, \dots, f_k$ of $\mathbb{Z}^k$ and integers $d_i \geq 1$ with $\Lambda = \bigoplus_i d_i\mathbb{Z}f_i$ (Smith normal form). Then $\mathbb{C}(\zeta) = \mathbb{C}(\zeta^{f_1}, \dots, \zeta^{f_k})$ is purely transcendental in the ζ^{f_i} and $\mathbb{C}(z) = \mathbb{C}(\zeta^{d_1 f_1}, \dots, \zeta^{d_k f_k})$, so $[\mathbb{C}(\zeta) : \mathbb{C}(z)] = \prod_i d_i = [\mathbb{Z}^k : \Lambda] = |G|$, and $\mathbb{C}(z) = \mathbb{C}(\zeta)^G$ by Artin's

theorem. The conjugates $\lambda(\zeta g)$, $g \in G$, of $\lambda(\zeta)$ are pairwise distinct, because the coefficient of ζ_j in $\lambda(\zeta g) - \lambda(\zeta g')$ is $g_j - g'_j$. Hence the minimal polynomial of $\lambda(\zeta)$ over $\mathbb{C}(z)$ has degree $|G|$ and equals F_G, which is monic with coefficients in $\mathbb{C}[\zeta^{\pm 1}]^G = \mathbb{C}[z^{\pm 1}]$; by Gauss's lemma it is irreducible in $\mathbb{C}[z^{\pm 1}][x]$. The invariance of $\det(x - M_W^G(\zeta))$ under $\zeta \mapsto \zeta g$ was noted above. □

3. Level-set degenerations

3.1. **The degeneration theorem.** For $J \subset [d]$, $\sigma \in \{\pm 1\}^J$ and $c \in (\mathbb{C}^*)^J$ put

$$\ell_{J,\sigma,c}(\theta) = \sum_{j \in J} c_j \theta_j^{\sigma_j} \qquad (\theta \in H).$$

For $s \in \mathbb{C}^J$ and $\xi \in (\mathbb{C}^*)^{J^c}$, $J^c = [d] \setminus J$, put

$$\Delta_{s,\xi}(\theta) = \sum_{j \in J} s_j \theta_j^{\sigma_j} + \sum_{j \in J^c} (\xi_j \theta_j + \xi_j^{-1} \theta_j^{-1}).$$

For $K \subset H$ write $C_K = (\widehat{V}(\theta' \theta^{-1}))_{\theta', \theta \in K}$.

Theorem 3.1 (Level-set degeneration)**.** *Let V be Floquet isospectral to 0 and let K be a nonempty level set of $\ell = \ell_{J,\sigma,c}$ in H. Then for all $s \in \mathbb{C}^J$ and $\xi \in (\mathbb{C}^*)^{J^c}$,*

$$\det(y - \operatorname{diag}(\Delta_{s,\xi})|_K - C_K) = \det(y - \operatorname{diag}(\Delta_{s,\xi})|_K) \qquad (y \in \mathbb{C}). \tag{3}$$

Proof. For large $t > 0$ define $\zeta(t)$ by $\zeta_j = (tc_j + s_j)^{\sigma_j}$ for $j \in J$ and $\zeta_j = \xi_j$ for $j \in J^c$. For $j \in J$ one has, for either sign, $\zeta_j \theta_j + \zeta_j^{-1} \theta_j^{-1} = (tc_j + s_j)\theta_j^{\sigma_j} + (tc_j + s_j)^{-1} \theta_j^{-\sigma_j}$. Hence

$$\lambda(\zeta(t)\theta) = t\,\ell(\theta) + \Delta_{s,\xi}(\theta) + r_t(\theta), \qquad r_t(\theta) = O(1/t).$$

Let ℓ_0 be the value of ℓ on K and let $A(t) = (t\ell_0 + y)I - M_V(\zeta(t))$. Its diagonal entries are $t(\ell_0 - \ell(\theta)) + O(1)$, with $\ell_0 - \ell(\theta) \neq 0$ for $\theta \notin K$, and $y - \Delta_{s,\xi}(\theta) - \widehat{V}(1) - r_t(\theta)$ for $\theta \in K$; its off-diagonal entries are the constants $-\widehat{V}(\theta' \theta^{-1})$. In the Leibniz expansion $\det A = \sum_\pi \operatorname{sgn}(\pi) \prod_\theta A_{\pi(\theta)\theta}$ the term of π is $O(t^{f(\pi)})$, uniformly for y in compact sets, where $f(\pi)$ is the number of $\theta \notin K$ fixed by π. Thus $f(\pi) \leq N - |K|$, with equality exactly when π fixes $H \setminus K$ pointwise. Summing the latter terms,

$$\det A(t) = t^{N-|K|} \Big(\kappa_K \det(y - \operatorname{diag}(\Delta_{s,\xi})|_K - C_K) + O(1/t) \Big), \qquad \kappa_K = \prod_{\theta \notin K} (\ell_0 - \ell(\theta)) \neq 0.$$

The same computation with $V = 0$ gives the right side of (3) in place of the left. By Lemma 2.2 the two determinants agree for every t; divide by $\kappa_K t^{N-|K|}$ and let $t \to \infty$. □

Lemma 3.2. *If K is a finite abelian group and $f \colon K \to \mathbb{C}$, the matrix $(f(\theta' \theta^{-1}))_{\theta', \theta \in K}$ is diagonalized by the characters of K with eigenvalues $\sum_k f(k)\chi(k)$. In particular it is nilpotent only if $f = 0$.*

Proof. Standard: $\sum_\theta f(\theta' \theta^{-1}) \chi(\theta)^{-1} = \chi(\theta')^{-1} \sum_k f(k)\chi(k)$; nilpotency forces every Fourier coefficient of f to vanish. □

3.2. **Cluster and crossing.**

Corollary 3.3 (Directional averages)**.** *Let V be Floquet isospectral to 0 and $j \in [d]$. The potential $V^{(j)}(n') = \sum_{h \in H_j} \widehat{V}(h)\,(\pi_{\hat{\jmath}} h)^{n'}$ on $\mathbb{Z}^{d-1}$, i.e. the average of V along e_j, is $\pi_{\hat{\jmath}}(\Gamma)$-periodic and Floquet isospectral to 0. In particular, if $\pi_{\hat{\jmath}}(\Gamma)$ is rigid then $\widehat{V}|_{H_j} = 0$.*

Proof. Take $J=\{j\}$ and $K=H_j$, the level set of $\ell=c_j\theta_j$ through 1, and $s=0$. The coordinate projection is injective on H_j and $\pi_{\hat\jmath}(H_j)=\pi_{\hat\jmath}(\Gamma)^\perp$: indeed $\theta'\in\pi_{\hat\jmath}(\Gamma)^\perp$ iff θ' with 1 inserted in position j lies in H. Thus (3) is the statement that $M^{\pi_{\hat\jmath}(\Gamma)}_{V^{(j)}}(\xi)$ and $M^{\pi_{\hat\jmath}(\Gamma)}_0(\xi)$ have the same characteristic polynomial for every $\xi\in(\mathbb{C}^*)^{d-1}$. □

Definition 3.4 (Balanced subgroups)**.** Let $J\subset[d]$, $\sigma\in\{\pm1\}^J$, and let P be a partition of J into blocks of size at least 2. Put

$$K_{J,\sigma,P}=\{\theta\in H:\ \theta_j^{\sigma_j}=\theta_k^{\sigma_k}\text{ whenever } j,k \text{ lie in the same block of } P\}.$$

An element $\eta\in H$ is *(J,σ,P)-balanced* if $\eta\in K_{J,\sigma,P}$, and *balanced* if it is $([d],\sigma,\{[d]\})$-balanced for some σ, i.e. $\eta_1^{\sigma_1}=\cdots=\eta_d^{\sigma_d}$.

Lemma 3.5. *$K=K_{J,\sigma,P}$ is a subgroup of H, and for generic c in $W_P=\{c\in\mathbb{C}^J:\sum_{j\in B}c_j=0$ for every block $B\}$ it is the level set of $\ell_{J,\sigma,c}$ through 1, with all $c_j\neq0$.*

Proof. K is the preimage of a subgroup under the homomorphism $\theta\mapsto(\theta_j^{\sigma_j})_{j\in J}$. For $\theta\in H$ put $w(\theta)=(\theta_j^{\sigma_j}-1)_{j\in J}$, so that $\ell_{J,\sigma,c}(\theta)-\ell_{J,\sigma,c}(1)=\langle c,w(\theta)\rangle$. The annihilator of W_P for this pairing is spanned by the block indicators, so $\langle c,w(\theta)\rangle=0$ for all $c\in W_P$ iff $w(\theta)$ is constant on every block, i.e. iff $\theta\in K$. For $\theta\notin K$ the functional $c\mapsto\langle c,w(\theta)\rangle$ is not identically zero on W_P. Since blocks have size ≥2, no coordinate functional vanishes on W_P. Avoiding finitely many proper hyperplanes of W_P gives the claim. □

Corollary 3.6 (Crossing)**.** *If V is Floquet isospectral to 0 and P is a partition of $[d]$ into blocks of size at least 2, then $\widehat V$ vanishes on $K_{[d],\sigma,P}$ for every σ. In particular $\widehat V(\eta)=0$ for every balanced $\eta\in H$.*

Proof. Theorem 3.1 with $J=[d]$, $s=0$ and Lemma 3.5 show that C_K is nilpotent, and Lemma 3.2 gives $\widehat V|_K=0$. For balanced η take $P=\{[d]\}$. □

3.3. **Fibration.**

Lemma 3.7 (Fibration)**.** *Let $K\subset H$ be a subgroup and $I\subset[d]$ a set of coordinates such that*

$$\det\Big(y-\operatorname{diag}\big(\textstyle\sum_{j\in I}\lambda_j(\xi_j\theta_j)\big)_{\theta\in K}-C_K\Big)=\det\Big(y-\operatorname{diag}\big(\textstyle\sum_{j\in I}\lambda_j(\xi_j\theta_j)\big)_{\theta\in K}\Big)\tag{4}$$

for all $\xi\in(\mathbb{C}^*)^I$, where $\lambda_j(w)=w+w^{-1}$. Put $K_0=\{\kappa\in K:\kappa_j=1\ (j\in I)\}$ and $G=\pi_I(K)\subset\mathbb{T}^I$. For a character ψ of K_0 choose a character Ψ of K extending ψ and define $F_\psi\colon G\to\mathbb{C}$ by

$$F_\psi(\pi_I k)=\Psi(k)^{-1}\sum_{\kappa\in K_0}\psi(\kappa)^{-1}\widehat V(k\kappa)\qquad(k\in K).$$

Then F_ψ is well defined and the $G^\perp$-periodic potential on $\mathbb{Z}^I$ with Fourier coefficients F_ψ is Floquet isospectral to 0. Consequently, if $G^\perp\subset\mathbb{Z}^I$ is rigid, then $\widehat V|_K=0$.

Proof. Write Δ_ξ for the diagonal operator in (4); it depends only on $\pi_I\theta$, hence is constant on K_0-cosets. For f on K write $f=\Psi g$. Then $f(\theta\kappa)=\psi(\kappa)f(\theta)$ for all $\kappa\in K_0$ iff g is K_0-invariant, so this isotypic subspace U_ψ is identified with functions on $K/K_0\cong G$. For $f=\Psi g\in U_\psi$,

$$(C_Kf)(\theta')=\sum_{k\in K}\widehat V(k)\Psi(\theta'k^{-1})g(\theta'k^{-1})=\Psi(\theta')\sum_{k_1\in K/K_0}\Big[\Psi(k_1)^{-1}\sum_{\kappa\in K_0}\psi(\kappa)^{-1}\widehat V(k_1\kappa)\Big]g(\theta'k_1^{-1}),$$

and $\Delta_\xi(\Psi g)=\Psi\Delta_\xi g$. Thus on U_ψ the operator $\Delta_\xi+C_K$ is, in the coordinates g, the Floquet matrix $M^G_{W_\psi}(\xi)$ of the potential W_ψ with Fourier coefficients F_ψ (Lemma 2.1 for $\mathbb{Z}^I$ and $G^\perp$). Well-definedness of F_ψ on G follows from the same computation. Summing over ψ, (4) becomes $\prod_\psi\det(y-M^G_{W_\psi}(\xi))=F_G(y,\xi)^{|K_0|}$. Each factor on the left is monic in y of degree $|G|$ with

coefficients in $\mathbb{C}[z^{\pm1}]$, and F_G is irreducible (Lemma 2.4); unique factorization in $\mathbb{C}[z^{\pm1}][y]$ forces every factor to equal F_G. If $G^\perp$ is rigid then every F_ψ vanishes, and Fourier inversion on K_0 gives $\widehat{V}(k\kappa) = 0$. □

Hypothesis (4) holds in two situations. First, if K is a level set of some $\ell_{J,\sigma,c}$ and $I = J^c$, by Theorem 3.1 with $s = 0$. Second, at *special points*: if $K = H$, $I \subset [d]$, and the coordinates $j \notin I$ can be frozen at values ζ_j^* such that $\lambda_j(\zeta_j^*\theta_j)$ does not depend on $\theta \in H$, then (4) is Lemma 2.2 at $\zeta_j = \zeta_j^*$. This happens when $\pi_j(H) \subset \{\pm1\}$, with $\zeta_j^* = i$.

3.4. **Single modes and pairs.**

Lemma 3.8 (Single mode)**.** *If V is Floquet isospectral to 0 and $\operatorname{supp}\widehat{V} \subset \{g\}$ for one $g \in H$, then $V = 0$.*

Proof. By (2) we may assume $g \neq 1$; let r be its order and $\alpha = \widehat{V}(g)$. Then $C_V = \alpha T_g$ permutes each coset of $\langle g\rangle$ cyclically, and for a diagonal X and an r-cycle P one has $\det(X - \alpha P) = \det X - \alpha^r$. Hence $P_V(x,\zeta) = \prod_{\text{cosets } Q}(\prod_{\theta\in Q}(x - \lambda(\zeta\theta)) - \alpha^r)$. The coefficient of x^{N-r} differs from that of P_0 by $-(N/r)\alpha^r$. □

Lemma 3.9 (Pairs)**.** *Let $h \in H$ have at least two coordinates different from 1, and let $\sigma \in \{\pm1\}^d$ be such that $w(h) = (h_j^{\sigma_j} - 1)_j$ is not proportional to $w(g)$ for any $g \in H \setminus \{1, h\}$. Then every isospectral V satisfies $\widehat{V}(h)\widehat{V}(h^{-1}) = 0$. If $h \in H^\circ$ is not balanced for σ (that is, the numbers $h_j^{\sigma_j}$ are not all equal), the hypothesis holds for this σ; in particular it holds for every σ if $h \in H^\circ$ is not balanced.*

Proof. Take $J = [d]$ and c generic in the hyperplane $\{\langle c, w(h)\rangle = 0\}$; it contains vectors with all $c_j \neq 0$ because $w(h)$ has two nonzero entries. As in Lemma 3.5, the level set through 1 is $\{\theta : w(\theta) \in \mathbb{C}w(h)\} = \{1, h\}$. By Theorem 3.1 and (2) the matrix $\begin{pmatrix} 0 & \widehat{V}(h^{-1}) \\ \widehat{V}(h) & 0 \end{pmatrix}$ is nilpotent. For the last statement, suppose $w(g) = s_0\, w(h)$ for some $s_0 \in \mathbb{C}$, so that $g^\sigma = (g_j^{\sigma_j})_j = \mathbf{1} + s_0\, w(h)$ lies on the complex line $\{\mathbf{1} + s\, w(h) : s \in \mathbb{C}\}$ and on $\mathbb{T}^d$. If $h \in H^\circ$ is not balanced for σ, there are j, k with $w_j(h) \neq w_k(h)$, both nonzero. The sets $\{s : |1 + s\, w_j(h)| = 1\}$ and $\{s : |1 + s\, w_k(h)| = 1\}$ are circles with the different centres $-1/w_j(h)$ and $-1/w_k(h)$, both pass through $s = 0$ and $s = 1$ (because $|1 + w_j(h)| = |h_j^{\sigma_j}| = 1$), and both contain s_0. Two distinct circles meet in at most two points, so $s_0 \in \{0, 1\}$ and $g \in \{1, h\}$. □

4. Rectangular periods

Theorem 4.1. *For every $d \geq 1$ the lattice $(3\mathbb{Z})^d$ is rigid.*

Proof. Here $H = \mu_3^d$, with $\mu_3 = \{1, \omega, \omega^2\}$. For $d = 1$ write $M = D + C$; then $\sum_\theta \lambda(\zeta\theta) = 0$, $C_{\theta\theta} = \widehat{V}(1) = 0$ and $(C^2)_{\theta\theta} = \sum_h \widehat{V}(h)\widehat{V}(h^{-1})$ is independent of θ. Comparing $\operatorname{tr} M^2$ and $\operatorname{tr} M^3$ with $\operatorname{tr} D^2$ and $\operatorname{tr} D^3$ gives $6\widehat{V}(\omega)\widehat{V}(\omega^2) = 0$ and $3(\widehat{V}(\omega)^3 + \widehat{V}(\omega^2)^3) = 0$, hence $V = 0$.

Let $d \geq 2$ and assume the claim in dimension $d - 1$. Since $\pi_j((3\mathbb{Z})^d) = (3\mathbb{Z})^{d-1}$, Corollary 3.3 gives $\widehat{V} = 0$ on $\bigcup_j H_j$. Let $\eta \in H^\circ = \{\omega, \omega^2\}^d$ and put $\sigma_j = 1$ if $\eta_j = \omega$ and $\sigma_j = -1$ if $\eta_j = \omega^2$. Then $\eta_j^{\sigma_j} = \omega$ for all j, so η is balanced and Corollary 3.6 gives $\widehat{V}(\eta) = 0$. □

Theorem 4.2. *$\Gamma = q_1\mathbb{Z} \oplus \cdots \oplus q_d\mathbb{Z}$ is rigid if and only if $\max_j q_j \leq 3$.*

Proof. If some $q_j \geq 4$ then $g_j(\Gamma) = q_j \geq 4$ and Proposition 5.1 below applies. Conversely let all $q_j \leq 3$; we induct on d. For $d = 1$: $q = 1$ is (2); for $q = 2$, $\det(x - M)$ differs from the free one by $-\widehat{V}(-1)^2$; $q = 3$ is Theorem 4.1. Let $d \geq 2$. Every projection $\pi_j(\Gamma)$ is rectangular with periods ≤ 3, so $\widehat{V} = 0$ on $\bigcup_j H_j$ by induction and Corollary 3.3. If some $q_j = 1$ then $H^\circ = \emptyset$. Otherwise

let $A = \{j : q_j = 3\}$ and $B = \{j : q_j = 2\}$, so $H = \mu_3^A \times \mu_2^B$, and let $\eta \in H^\circ$; then $\eta_B = -\mathbf{1}_B$ and $\eta_A \in \{\omega, \omega^2\}^A$.

Case $|A| \geq 2$. Choose σ on A with $\eta_j^{\sigma_j} = \omega$ $(j \in A)$ and let $K = K_{A,\sigma,\{A\}} = \langle \eta_A \rangle \times \mu_2^B$. By Lemma 3.5, Theorem 3.1 and Lemma 3.7 with $I = B$, the fibred lattice is $\pi_B(K)^\perp = (2\mathbb{Z})^B$, rigid by induction (or vacuous if $B = \emptyset$). Hence $\widehat{V}|_K = 0$ and $\widehat{V}(\eta) = 0$.

Case $|A| \leq 1$, $|B| \geq 2$. Let $K = K_{B,\sigma,\{B\}} = \mu_3^A \times \langle -\mathbf{1}_B \rangle \ni \eta$ (any σ). Lemma 3.7 with $I = A$ gives the fibred lattice $(3\mathbb{Z})^A$, which is rigid; hence $\widehat{V}(\eta) = 0$.

Case $|A| = |B| = 1$, i.e. $\Gamma = 3\mathbb{Z} \oplus 2\mathbb{Z}$ up to order. Here $H^\circ = \{\eta, \eta^{-1}\}$ with $\eta = (\omega, -1)$, which is not balanced, so Lemma 3.9 gives $\widehat{V}(\eta)\widehat{V}(\eta^{-1}) = 0$ and Lemma 3.8 gives $V = 0$. □

Remark 4.3. The proof uses no perturbation series, no pole separation and no symbolic computation. CFK obtained the rigid cases with at most two periods equal to 3 from symbolic computation in dimension ≤ 2 and a dimension reduction for periods 1 and 2 [3, Proposition 4.2], and noted that they could not find a reduction for period 3 [3, Section 4.4]. A reduction of this kind was first given in [8, Part 12] by perturbative means; the crossing step of Theorem 4.1 is a non-perturbative one.

5. Arbitrary periods

5.1. **Nonrigidity.**

Proposition 5.1. *If $g_j(\Gamma) \geq 4$ for some j, then Γ is nonrigid. More precisely, if $v\colon \mathbb{Z} \to \mathbb{C}$ is $g_j(\Gamma)$-periodic and Floquet isospectral to 0, then $V(n) = v(n_j)$ is Γ-periodic and Floquet isospectral to 0.*

Proof. Nonzero q-periodic one-dimensional potentials isospectral to 0 exist for every $q \geq 4$ [3, Section 4.3 and Theorem C]. Since $\pi_j(\Gamma) = g_j\mathbb{Z}$, V is Γ-periodic, with $\operatorname{supp} \widehat{V} \subset A_j$. Then $M_V(\zeta)$ is block diagonal along the cosets $\theta_0 A_j$, and the block of $\theta_0 A_j$ is $(\sum_{k \neq j} \lambda_k(\zeta_k \theta_{0,k}))I + M_v^{g_j\mathbb{Z}}(\zeta_j \theta_{0,j})$, which is isospectral to its free counterpart. □

5.2. **Splitting off $2\mathbb{Z}$.**

Theorem 5.2. *For every full-rank lattice $\Gamma' \subset \mathbb{Z}^{d-1}$, the lattice $\Gamma' \oplus 2\mathbb{Z} \subset \mathbb{Z}^d$ is rigid if and only if Γ' is rigid.*

Proof. If W is a nonzero Γ'-periodic potential isospectral to 0, then $V(n', n_d) = W(n')$ is nonzero, $\Gamma' \oplus 2\mathbb{Z}$-periodic and isospectral. Conversely let Γ' be rigid and V isospectral for $\Gamma = \Gamma' \oplus 2\mathbb{Z}$, so $H = H' \times \{\pm 1\}$. Corollary 3.3 with $j = d$ gives $\widehat{V}(\cdot, 1) = 0$. Freeze $\zeta_d = i$; then $\lambda_d(i\theta_d) = 0$ for $\theta_d = \pm 1$, and Lemma 3.7 applies to $K = H$, $I = [d-1]$, $K_0 = \{1\} \times \{\pm 1\}$, $G = H'$. With $\Psi(\theta', \theta_d) = \psi(\theta_d)$ the fibred potentials are $F_\pm = \pm\widehat{V}(\cdot, -1)$, so $\widehat{V}(\cdot, -1)$ is the Fourier transform of a Γ'-periodic isospectral potential and vanishes. □

This extends the period-2 case of [3, Proposition 4.2] from rectangular to arbitrary Γ'. The analogous statement for $\Gamma' \oplus 3\mathbb{Z}$ holds for rectangular Γ' (Theorem 4.2) and for the diagonal lattices of Example 5.6 (Example 5.7), and is open in general (Section 8).

Corollary 5.3. *If $m_j(\Gamma) \leq 3$ for all j, then Γ is rigid.*

Proof. $\Gamma \supset m_1\mathbb{Z} \oplus \cdots \oplus m_d\mathbb{Z}$; apply Lemma 2.3 and Theorem 4.2. □

5.3. **A recursive criterion.**

Definition 5.4. Let $d \geq 2$ and $\Gamma \subset \mathbb{Z}^d$. An element $\eta \in H^\circ$ is *resolved* if it lies in a balanced subgroup $K = K_{J,\sigma,P}$ (Definition 3.4) whose *fibred lattice* $\pi_{J^c}(K)^\perp \subset \mathbb{Z}^{J^c}$ is rigid; for $J = [d]$ the fibred lattice is the zero-dimensional lattice, which we regard as rigid.

Theorem 5.5. *Let $d \geq 2$ and suppose that every projection $\pi_j(\Gamma)$ is rigid. If every element of H° is resolved, except possibly the two elements of one pair $\{h, h^{-1}\}$, then Γ is rigid.*

Proof. By Corollary 3.3, $\operatorname{supp} \widehat{V} \subset H^\circ$. For a resolved η with $K = K_{J,\sigma,P}$, Lemma 3.5, Theorem 3.1 with $s = 0$, and Lemma 3.7 with $I = J^c$ give $\widehat{V}|_K = 0$ (for $J = [d]$ use Corollary 3.6). The remaining support lies in $\{h, h^{-1}\}$. If h were balanced for some σ, then h and h^{-1}, which lie in the subgroup $K_{[d],\sigma,\{[d]\}}$, would be resolved (take $J = [d]$ and $P = \{[d]\}$) and the remaining support would be empty. So we may assume that h is not balanced; then by Lemma 3.9 at most one of $\widehat{V}(h)$, $\widehat{V}(h^{-1})$ is nonzero, and Lemma 3.8 finishes. □

Example 5.6 (Diagonal lattices)**.** Let $\varepsilon \in \{\pm 1\}^d$, $d \geq 2$, $N \geq 1$, and $\Gamma_{\varepsilon,N} = \{n \in \mathbb{Z}^d : \sum_j \varepsilon_j n_j \equiv 0 \pmod N\}$. Then $H = \langle (u^{\varepsilon_1}, \dots, u^{\varepsilon_d}) \rangle$ with $u = e^{2\pi i/N}$, every element of H is balanced (take $\sigma = \varepsilon$), and $\pi_j(\Gamma_{\varepsilon,N}) = \mathbb{Z}^{d-1}$. Hence $\Gamma_{\varepsilon,N}$ is rigid. For $\varepsilon = (1, -1, \dots, -1)$ this is [4, Theorem 6.1]; the general sign pattern is equivalent to it under the reflections $n_j \mapsto -n_j$, which preserve Δ.

Example 5.7 (Splitting off $3\mathbb{Z}$ from diagonal lattices)**.** Let $d \geq 3$, $\varepsilon \in \{\pm 1\}^{d-1}$, $N \geq 1$, and $\Gamma = \Gamma_{\varepsilon,N} \oplus 3\mathbb{Z} \subset \mathbb{Z}^d$ with $\Gamma_{\varepsilon,N} \subset \mathbb{Z}^{d-1}$ as in Example 5.6. Then $H = H' \times \mu_3$ with $H' = \langle (u^{\varepsilon_1}, \dots, u^{\varepsilon_{d-1}}) \rangle$, the directional periods are $(N, \dots, N, 3)$ and the axis indices are $(1, \dots, 1, 3)$. Every element of H lies in $K = K_{[d-1],\varepsilon,\{[d-1]\}}$, so $K = H$, and its fibred lattice $\pi_{\{d\}}(K)^\perp = 3\mathbb{Z} \subset \mathbb{Z}^{\{d\}}$ is rigid. The projections are rigid: $\pi_{\hat{d}}(\Gamma) = \Gamma_{\varepsilon,N}$ (Example 5.6), and $\pi_{\hat{j}}(\Gamma) = \mathbb{Z}^{d-2} \oplus 3\mathbb{Z}$ for $j < d$ (Theorem 4.2). By Theorem 5.5, Γ is rigid. For $d = 3$ and $\gcd(N, 3) = 1$ the Chinese remainder theorem identifies Γ with $\{n : 3(\varepsilon_1 n_1 + \varepsilon_2 n_2) + N n_3 \equiv 0 \pmod{3N}\}$, whose dual group is cyclic of order $3N$. For $N \geq 4$ these lattices are neither rectangular nor covered by Corollary 5.3. The analogous statement with $2\mathbb{Z}$ in place of $3\mathbb{Z}$ is a special case of Theorem 5.2 and Example 5.6.

Example 5.8 (One unbalanced pair)**.** Let $\Gamma = \{n \in \mathbb{Z}^2 : n_1 + 2n_2 \equiv 0 \pmod 6\} = (6,0)\mathbb{Z} \oplus (4,1)\mathbb{Z}$, with axis indices $(2, 1)$ and directional periods $(6, 3)$. With $u = e^{\pi i/3}$, $H = \{(u^k, u^{2k})\}$. The elements with $k = 2, 4$ are balanced for $\sigma = (1, -1)$, the element with $k = 3$ lies on an axis, and $\{k = 1, k = 5\}$ is a single unbalanced pair. Theorem 5.5 gives rigidity.

5.4. **Reduction to off-axis vanishing.**

Theorem 5.9. *For $d \geq 2$ consider the statement*

> (OV_d) *If $\Gamma \subset \mathbb{Z}^d$ satisfies $\max_j g_j(\Gamma) \leq 3$ and V is Floquet isospectral to 0 with $\operatorname{supp} \widehat{V} \subset H^\circ$, then $V = 0$.*

Conjecture 1.2 holds for all lattices in $\mathbb{Z}^d$ and all $d \leq D$ if and only if (OV_d) holds for $2 \leq d \leq D$.

Proof. Necessity is clear. For sufficiency, the case $d = 1$ holds because $g_1(\Gamma) = [\mathbb{Z} : \Gamma]$, by Theorem 4.2 and [3]. Let $d \geq 2$ and $\max_j g_j(\Gamma) \leq 3$. Since $g_k(\pi_{\hat{j}}(\Gamma)) = g_k(\Gamma)$ for $k \neq j$, the projections satisfy the hypothesis in dimension $d - 1$ and are rigid by induction; by Corollary 3.3, $\operatorname{supp} \widehat{V} \subset H^\circ$, and ($\mathrm{OV}_d$) gives $V = 0$. The converse direction of the conjecture is Proposition 5.1. □

Remark 5.10. Comparing the traces of the squares of $M_V(\zeta)$ and $M_0(\zeta)$ gives $\sum_{h \in H} \widehat{V}(h)\widehat{V}(h^{-1}) = 0$ for every isospectral V. For real potentials $\widehat{V}(h^{-1}) = \overline{\widehat{V}(h)}$, so this identity alone gives $V = 0$. For complex potentials Lemma 3.9 gives the termwise identities $\widehat{V}(h)\widehat{V}(h^{-1}) = 0$ for unbalanced $h \in H^\circ$, and Corollary 3.6 gives $\widehat{V}(h) = 0$ for balanced h; so $\widehat{V}(h)\widehat{V}(h^{-1}) = 0$ for every $h \in H^\circ$, in every dimension $d \geq 2$. These identities say only that the support meets each pair $\{h, h^{-1}\} \subset H^\circ$ in at most one element, and higher-order information is needed.

We now describe the first higher-order identities. Let $d = 2$, let V be isospectral with $S = \operatorname{supp} \widehat{V} \subset H^\circ$ nonempty, and assume (by Corollary 3.6 and Lemma 3.9) that S contains no balanced

element and $S \cap S^{-1} = \emptyset$. A *cycle* of length m is a word $(s_1, \dots, s_m) \in S^m$ with $s_1 \cdots s_m = 1$; its vertices are the partial products $p_i = s_1 \cdots s_i$, $1 \le i < m$. Let m_0 be the minimal length of a cycle (it exists since H is finite), and let $\mathfrak{M}$ be the set of multisets $\{s_1, \dots, s_{m_0}\}$ of minimal cycles. For a chart $\sigma \in \{\pm 1\}^2$ and $c \in (\mathbb{C}^*)^2$ put $L_p^\sigma(c) = \sum_j c_j(1 - p_j^{\sigma_j})$ and, for $M \in \mathfrak{M}$,

$$R_M^\sigma(c) = \sum_{(s_1,\dots,s_{m_0})} \prod_{i=1}^{m_0-1} \frac{1}{L^\sigma_{s_1\cdots s_i}(c)},$$

the sum running over the distinct orderings of M.

Proposition 5.11 (Minimal-cycle identities)**.** *In this situation, for each chart σ,*

$$\sum_{M\in\mathfrak{M}} \Big(\prod_{s\in M} \widehat{V}(s)\Big) R_M^\sigma(c) = 0 \qquad \text{identically in } c. \tag{5}$$

Moreover each $s \in S$ occurs as a vertex of a minimal cycle only as the first vertex, and only in cycles whose multiset contains s. Consequently, if some $s \in S$ lies in exactly one $M \in \mathfrak{M}$ and the residue of R_M^σ along $L_s^\sigma = 0$ is nonzero for some σ, we obtain a contradiction; under that hypothesis $S = \emptyset$.

Proof. Let $\mathcal{G} = H \setminus \{1\}$, $\Lambda_p(\zeta) = \lambda(\zeta) - \lambda(\zeta p)$, and $C_\mathcal{G} = (\widehat{V}(pq^{-1}))_{p,q\in\mathcal{G}}$. Since $\lambda(\zeta)$ is an eigenvalue of $M_V(\zeta)$ (Lemma 2.2), the Schur complement at the index 1 gives, wherever $\det(\Lambda - C_\mathcal{G}) \neq 0$,

$$0 = \det(\lambda(\zeta) - M_V(\zeta)) = -\det(\Lambda - C_\mathcal{G})\Big(\widehat{V}(1) + \tilde{v}^{\,T}(\Lambda - C_\mathcal{G})^{-1} v\Big),$$

where $v = (\widehat{V}(q))_{q\in\mathcal{G}}$ and $\tilde{v} = (\widehat{V}(p^{-1}))_{p\in\mathcal{G}}$. In the chart $\zeta_j = (tc_j)^{\sigma_j}$ one has $\Lambda_p = tL_p^\sigma(c) + t^{-1}L_p'(c)$ with $L_p'(c) = \sum_j c_j^{-1}(1 - p_j^{-\sigma_j})$, so for large t the Neumann series $\tilde{v}^T(\Lambda - C_\mathcal{G})^{-1}v = \sum_{k\ge 0} \tilde{v}^T(\Lambda^{-1}C_\mathcal{G})^k\Lambda^{-1}v$ converges. The k-th term is a sum over closed walks of length $k+2$ from 1, with steps in S, and is $O(t^{-k-1})$. Since there are no cycles of length $< m_0$, all terms with $k < m_0 - 2$ vanish, and the coefficient of $t^{-(m_0-1)}$ is the left side of (5). (Minimal cycles have pairwise distinct vertices, all different from 1.)

If $s \in S$ were the i-th vertex of a minimal cycle with $i \ge 2$, then $(s, s_{i+1}, \dots, s_{m_0})$ would be a shorter cycle. So L_s^σ appears in (5) only through orderings starting with s of multisets containing s. By Lemma 3.9 no other vertex p has L_p^σ proportional to L_s^σ. Taking the residue along $L_s^\sigma = 0$ gives $\widehat{V}(s)\prod_{r\in M\setminus\{s\}} \widehat{V}(r) \cdot \operatorname{Res} R_M^\sigma = 0$, which is impossible when all factors are nonzero. □

Remark 5.12. The hypothesis of the last sentence is satisfied in many cases (for instance whenever $\mathfrak{M}$ has one element and some residue is nonzero), and it is easy to check by machine. It is not always satisfied: for the lattices $\{n : n_1 + an_2 \equiv 0 \pmod{31}\}$ with $a \not\equiv 0, \pm 1$, identifying H with $\mathbb{Z}/31$ via $k \mapsto (u^k, u^{ak})$, the support $S = \{8, \dots, 13\}$ has every element in at least two minimal multisets. A general argument covering such supports would prove (OV_2).

Example 5.13 (The smallest critical planar lattices)**.** Let $\Gamma = \{n \in \mathbb{Z}^2 : n_1 + 2n_2 \equiv 0 \pmod 5\} = (5,0)\mathbb{Z} \oplus (3,1)\mathbb{Z}$, so that $H = \{(u^k, u^{2k})\}$ with $u = e^{2\pi i/5}$; the axis indices are $(1,1)$ and the directional periods are $(5,5)$. Write k for (u^k, u^{2k}). No element of $H \setminus \{1\} = H^\circ$ is balanced, and H° consists of the two pairs $\{1,4\}$ and $\{2,3\}$, so Theorem 5.5 does not apply. Nevertheless Γ is rigid. Let V be isospectral and $S = \operatorname{supp} \widehat{V}$. By Lemmas 3.9 and 3.8 we may assume that S consists of one element of each pair, so $S \cap S^{-1} = \emptyset$. The reflection $V(n) \mapsto V(-n)$ preserves isospectrality, because $P_{V(-\cdot)}(x,\zeta) = P_V(x,\zeta^{-1})$, and replaces S by S^{-1}; so we may assume $S = \{1,2\}$ or $S = \{1,3\}$. Write $M_V(\zeta) = D + C$ with D diagonal and $C = (\widehat{V}(\theta'\theta^{-1}))$. Since $\widehat{V}(1) = 0$ and $\widehat{V}(h)\widehat{V}(h^{-1}) = 0$ for all h, the traces $\operatorname{tr}(D^2C)$ and $\operatorname{tr}(DC^2)$ vanish, and the equality of $\operatorname{tr} M_V(\zeta)^3$ and $\operatorname{tr} M_0(\zeta)^3$ gives $\operatorname{tr} C^3 = 5\sum_{h_1h_2h_3=1} \widehat{V}(h_1)\widehat{V}(h_2)\widehat{V}(h_3) = 0$. For $S = \{1,2\}$ the only solutions

of $k_1+k_2+k_3\equiv 0 \pmod 5$ in S are the three orderings of $(1,2,2)$, so $\operatorname{tr} C^3 = 15\,\widehat{V}(1)\widehat{V}(2)^2$; for $S=\{1,3\}$ they are the orderings of $(1,1,3)$ and $\operatorname{tr} C^3 = 15\,\widehat{V}(1)^2\widehat{V}(3)$. Both are nonzero, a contradiction. Exchanging coordinates gives the same conclusion for $(5,0)\mathbb{Z}\oplus(2,1)\mathbb{Z}$. These are the two planar lattices of smallest index to which neither Theorem 5.5 nor Corollary 5.3 applies.

6. Node coordinates in dimension one

In this section $d=1$, $q=2m+1\geq 3$ is odd, $\Gamma=q\mathbb{Z}$, $V_n=V(n)$, and ω denotes $e^{2\pi i/q}$. The Floquet determinant of $\Delta+V$ is $D_V(x)-z-z^{-1}$ with the discriminant $D_V(x)=\operatorname{tr}(T_q(x)\cdots T_1(x))$, $T_n(x)=\left(\begin{smallmatrix} x-V_n & -1\\ 1 & 0\end{smallmatrix}\right)$, so V is Floquet isospectral to 0 if and only if $D_V=D_0$; recall $D_0(t+t^{-1})=t^q+t^{-q}$. Since D_V is invariant under cyclic shifts of V, Conjecture 2 of [4] asks for a nonzero such V one of whose cyclic shifts has the form $(v_1,v_2,\dots,\overline{v_2},\overline{v_1})$. For even q it is proved in [4]; for odd q it is open. We give an explicit construction of such potentials from the node data of the free spectral curve.

Put $\theta_l=e^{\pi i l/q}$ for $1\leq l\leq 2m$, so that $\theta_l^2=\omega^l$. The two points $\theta_l^{\pm 1}$ of the normalization $\mathbb{C}^*\ni t$ of the free curve $\{D_0(x)=z+z^{-1}\}$, $x=t+t^{-1}$, $z=t^q$, are glued to its l-th double point. For $I\subset[2m]$ write $s(I)=\sum_{l\in I} l$ and $\beta^I=\prod_{l\in I}\beta_l$, and put

$$a_{lk}=\Big(\frac{\theta_k-\theta_l}{\theta_k\theta_l-1}\Big)^2=\Big(\frac{\sin(\pi(k-l)/2q)}{\sin(\pi(k+l)/2q)}\Big)^2\in(0,1),\qquad A_I=\prod_{l<k,\ l,k\in I} a_{lk}.$$

(The second expression follows from $\theta_k-\theta_l=2ie^{\pi i(k+l)/2q}\sin(\pi(k-l)/2q)$ and $\theta_k\theta_l-1=2ie^{\pi i(k+l)/2q}\sin(\pi(k+l)/2q)$, and $0<a_{lk}<1$ because $|k-l|<\min(k+l,2q-k-l)$ for $1\leq l\neq k\leq 2m$.) The sums $\sum_I A_I\beta^I\omega^{-ns(I)}$ below have the form of the Hirota N-soliton τ-functions of the Toda lattice at zero background, with spectral parameters at the double points (compare [14, 13]).

For $\varkappa\in\mathbb{C}^{2m}$ and $n\in\mathbb{Z}$ let $\mathcal{M}_n(\varkappa)=\big(\omega^{ln}\theta_l^j-\varkappa_l\theta_l^{-j}\big)_{1\leq l\leq 2m,\ 0\leq j\leq 2m-1}$ and $\tau_n(\varkappa)=\det\mathcal{M}_n(\varkappa)$.

Lemma 6.1 (Hirota form). *With $W_0=\det(\theta_l^j)\neq 0$, $\Pi_l=\prod_{k\neq l}\frac{\theta_k-\theta_l^{-1}}{\theta_k-\theta_l}\neq 0$ and $\beta_l=-\varkappa_l\Pi_l$,*

$$\tau_n(\varkappa)=W_0\sum_{I\subset[2m]} A_I\,\beta^I\,\omega^{-ns(I)}.$$

Proof. Expanding every row of $\mathcal{M}_n$ as the sum of $\omega^{ln}(\theta_l^j)_j$ and $-\varkappa_l(\theta_l^{-j})_j$ gives

$$\tau_n=\sum_I\prod_{l\in I}(-\varkappa_l)\prod_{l\notin I}\omega^{ln}\,W(I),$$

where $W(I)=\det(x_l^j)$ with $x_l=\theta_l^{-1}$ for $l\in I$ and $x_l=\theta_l$ otherwise. Since $s([2m])=q(q-1)/2\equiv 0 \pmod q$, the factor $\prod_{l\notin I}\omega^{ln}$ equals $\omega^{-ns(I)}$. By the Vandermonde formula $W(I)/W_0=\prod_{l<k}(x_k-x_l)/(\theta_k-\theta_l)$; a pair with exactly one index l in I contributes $F(l,k)=(\theta_k-\theta_l^{-1})/(\theta_k-\theta_l)$, a pair inside I contributes $-(\theta_l\theta_k)^{-1}$, and other pairs contribute 1. As $\Pi_l=\prod_{k\neq l}F(l,k)$, this gives

$$\frac{W(I)}{W_0}=\prod_{l\in I}\Pi_l\cdot\prod_{l<k,\ l,k\in I}\frac{-1}{\theta_l\theta_k\,F(l,k)F(k,l)},$$

and since $F(l,k)F(k,l)=-(\theta_l\theta_k-1)^2/(\theta_l\theta_k(\theta_k-\theta_l)^2)$ the last factor equals a_{lk}. The numbers Π_l are nonzero because θ_l^{-1} lies in the lower half-plane and the θ_k in the upper one. □

Theorem 6.2 (Node coordinates). *Let $\varkappa\in\mathbb{C}^{2m}$, $\nu_0\in\mathbb{Z}/q$ and $C\neq 0$ satisfy $\tau_n(\varkappa)=C\omega^{-n\nu_0}$ for all $n\in\mathbb{Z}$. For each n let P_n be the unique monic polynomial of degree $2m$ with*

$$\omega^{ln}P_n(\theta_l)=\varkappa_l P_n(\theta_l^{-1})\qquad(1\leq l\leq 2m),\tag{6}$$

and let p_n be its coefficient of t^{2m-1}. Then $V_n = p_n - p_{n+1}$ is q-periodic and Floquet isospectral to 0. Moreover:

(a) *$V = 0$ if and only if $\varkappa = 0$;*
(b) *with β as in Lemma 6.1 and $\delta(I) = -2i\sum_{l\in I}\sin(\pi l/q)$,*

$$V_n = \rho_{n+1} - \rho_n, \qquad \rho_n = \frac{\sum_I A_I \beta^I \delta(I)\,\omega^{-ns(I)}}{\sum_I A_I \beta^I\,\omega^{-ns(I)}};$$

(c) *if $\beta \in \mathbb{R}^{2m}$, then $V_{-1-n} = \overline{V_n}$ for all n; hence the shifted potential $U_n = V_{n-1}$, which is also Floquet isospectral to 0, satisfies $(U_1,\dots,U_q) = (V_0,\dots,V_{q-1}) = (v_1, v_2, \dots, \overline{v_2}, \overline{v_1})$.*

Proof. The coefficient vector c of $P_n - t^{2m}$ solves $\mathcal{M}_n(\varkappa)c = -b_n$ with $(b_n)_l = \omega^{ln}\theta_l^{2m} - \varkappa_l\theta_l^{-2m}$, so P_n exists and is unique because $\tau_n \neq 0$; since $\mathcal{M}_{n+q} = \mathcal{M}_n$, $P_{n+q} = P_n$ and V is q-periodic. Put $\varphi_n(t) = t^n P_n(t)$. As $\theta_l^{2n} = \omega^{ln}$, condition (6) reads $\varphi_n(\theta_l) = \varkappa_l\varphi_n(\theta_l^{-1})$, a relation whose form does not depend on n.

Constant terms. By Cramer's rule, $\tau_n P_n(0)$ is the determinant of $\mathcal{M}_n$ with its column $j = 0$ replaced by $-b_n$. Moving that column to the end and using that $2m$ is even, $\tau_n P_n(0) = \det(\omega^{ln}\theta_l^j - \varkappa_l\theta_l^{-j})_{1\le j\le 2m}$. Taking $\theta_l\omega^{-l}$ out of row l (recall $\theta_l^{-2} = \omega^{-l}$) turns the right side into $\prod_l \theta_l\,\omega^{-s([2m])}\tau_{n+1} = (-1)^m\tau_{n+1}$, because $\prod_l\theta_l = e^{\pi i(q-1)/2}$. Hence $P_n(0) = (-1)^m\tau_{n+1}/\tau_n = (-1)^m\omega^{-\nu_0}$ does not depend on n.

Three-term relation. Fix n and let $\Omega(t) = \varphi_{n+1}(t) + \varphi_{n-1}(t) + V_n\varphi_n(t) - (t+t^{-1})\varphi_n(t) = t^n S(t)$ with $S(t) = t(P_{n+1} - P_n) + t^{-1}(P_{n-1} - P_n) + V_nP_n$. The polynomial $tS(t)$ has degree at most $2m+1$, its coefficient of t^{2m+1} is $p_{n+1} - p_n + V_n = 0$, and its constant term is $P_{n-1}(0) - P_n(0) = 0$. So S is a polynomial of degree at most $2m-1$. Each φ_k satisfies the node relations and $t + t^{-1}$ takes the same value at $\theta_l^{\pm1}$, so $\Omega(\theta_l) = \varkappa_l\Omega(\theta_l^{-1})$, that is, $\omega^{ln}S(\theta_l) = \varkappa_l S(\theta_l^{-1})$ for all l. This says $\mathcal{M}_n(\varkappa)s = 0$ for the coefficient vector s of S, so $S = 0$.

Isospectrality. Thus for every $t \in \mathbb{C}^*$ the sequence $\varphi_n(t)$ solves $(\Delta + V)\varphi = (t + t^{-1})\varphi$, and $\varphi_{n+q}(t) = t^q\varphi_n(t)$. The polynomials $\varphi_0 = P_0$ and $\varphi_1 = tP_1$ have at most $2m$ common zeros, so for all other t the vector $(\varphi_1(t), \varphi_0(t))$ is a nonzero eigenvector, with eigenvalue t^q, of the monodromy matrix $T_q\cdots T_1$ at $x = t + t^{-1}$. Its determinant is 1, so $D_V(t + t^{-1}) = t^q + t^{-q} = D_0(t + t^{-1})$ for all but finitely many t, and $D_V = D_0$.

(a) If $\varkappa = 0$, then $P_n = \prod_l(t - \theta_l)$ for all n and $V = 0$. Conversely, let $V = 0$. For $t^{2q} \neq 1$ the free solutions with Floquet multiplier t^q are the multiples of $(t^n)_n$, so $P_n(t)$ does not depend on n, and $P_n = P_0$. Comparing (6) for n and $n+1$ gives $(\omega^l - 1)\omega^{ln}P_0(\theta_l) = 0$, so $P_0 = \prod_l(t - \theta_l)$; as θ_l^{-1} lies in the lower half-plane, $P_0(\theta_l^{-1}) \neq 0$, and (6) forces $\varkappa_l = 0$.

(b) By Cramer's rule $\tau_n p_n$ is the determinant of $\mathcal{M}_n$ with its column $j = 2m-1$ replaced by $-b_n$. Expanding rows as in Lemma 6.1, the term indexed by I is

$$-\prod_{l\in I}(-\varkappa_l)\,\omega^{-ns(I)}\det(x_l^0,\dots,x_l^{2m-2},x_l^{2m}),$$

and $\det(x_l^0,\dots,x_l^{2m-2},x_l^{2m}) = W(I)\,e_1(x)$ with $e_1(x) = \sum_l x_l = e_1(\theta) + \delta(I)$. Hence $p_n = -e_1(\theta) - \rho_n$, and $V_n = \rho_{n+1} - \rho_n$.

(c) If β is real, then $\overline{\sum_I A_I\beta^I\omega^{-ns(I)}} = \sum_I A_I\beta^I\omega^{ns(I)}$ and, since $\delta(I) \in i\mathbb{R}$, $\overline{\rho_n} = -\rho_{-n}$. Therefore $\overline{V_n} = \rho_{-n} - \rho_{-n-1} = V_{-1-n}$, and $U_{q+1-j} = V_{-j} = \overline{V_{j-1}} = \overline{U_j}$. □

Writing $E_\nu(\beta) = \sum_{s(I)\equiv\nu} A_I\beta^I$ for $\nu \in \mathbb{Z}/q$, Lemma 6.1 gives $\tau_n = W_0\sum_{\nu\in\mathbb{Z}/q}E_\nu(\beta)\omega^{-n\nu}$, so the hypothesis of Theorem 6.2 says that the vector $(E_\nu)_\nu$ is supported at the single residue ν_0. Since $\Pi_l \neq 0$, every $\beta \in \mathbb{C}^{2m}$ comes from a unique $\varkappa$.

Corollary 6.3 (Reduction of Conjecture 2)**.** *Let $q > 3$ be odd. Conjecture 2 of* [4] *holds for the period q if there are $\beta \in \mathbb{R}^{2m} \setminus \{0\}$ and $\nu_0 \in \mathbb{Z}/q$ such that*

$$E_\nu(\beta) = 0 \quad (\nu \neq \nu_0), \qquad E_{\nu_0}(\beta) \neq 0. \tag{7}$$

The equations form a square system of $q-1$ real multilinear equations in $q-1$ real unknowns with positive coefficients A_I. Every real solution of (7) *with $\beta \neq 0$ has a negative coordinate.*

Proof. The first statement is Theorem 6.2. If $\beta \geq 0$ and $\beta_l > 0$, then $E_l(\beta) \geq A_{\{l\}}\beta_l > 0$, so $l \equiv \nu_0$; if $\nu_0 \neq 0$ then $E_0(\beta) \geq A_\emptyset = 1 > 0$ is a contradiction, and if $\nu_0 = 0$ there is no such l. □

Remark 6.4 (Symmetries). If V is Floquet isospectral to 0, so are $-V$ (because $D_{-V}(x) = -D_V(-x)$ for odd q) and $\overline{V}$, and these maps preserve the conjugate-reflection form. For $V \neq 0$ of this form the four potentials $\pm V, \pm\overline{V}$ are distinct: comparing the coefficients of x^{q-1} and x^{q-2} of $\det(x - C_q - \operatorname{diag} V)$ and $\det(x - C_q)$ gives $\sum_n V_n = \sum_n V_n^2 = 0$, which excludes real and purely imaginary $V \neq 0$. Hence the number of nonzero solutions of the real system of Proposition 7.3, when it is finite, is divisible by 4. In node coordinates $-V$ is obtained as follows. Suppose that all $\varkappa_l \neq 0$ and put $\tilde{\varkappa}_{q-l} = 1/\varkappa_l$. Since $\theta_{q-l} = -\theta_l^{-1}$ and $\omega^{(q-l)n} = \omega^{-ln}$, a polynomial S satisfies the relations (6) for $\tilde{\varkappa}$ if and only if $S(-t)$ satisfies them for $\varkappa$. Hence $\tau_n(\tilde{\varkappa}) \neq 0$, the node polynomials of $\tilde{\varkappa}$ are $\tilde{P}_n(t) = P_n(-t)$, whose coefficient of t^{2m-1} is $-p_n$, and $\tilde{P}_n(0) = P_n(0)$. The identity $\tau_n P_n(0) = (-1)^m \tau_{n+1}$ from the proof of Theorem 6.2, which holds whenever $\tau_n \neq 0$, then shows that $\tilde{\varkappa}$ satisfies the hypothesis of Theorem 6.2 with the same ν_0, and the potential it produces is $-V$.

Remark 6.5. For $q = 5$ the real solution

$$\beta = (-2.111953\ldots,\ 4.663756\ldots,\ 1.587761\ldots,\ 1.232278\ldots)$$

of (7), with $\nu_0 = 2$ and $E_2 = 4.506505\ldots$, gives via Theorem 6.2(b) exactly the potential $(V_0, \ldots, V_4) = (v_1, v_2, v_3, \overline{v_2}, \overline{v_1})$ of Proposition 7.2. Conversely, for each of the 4, 115, 397 and 251 nonzero conjugate-reflection solutions returned by our Newton search for $q = 5, 7, 9, 11$ (before closing them under the symmetries of Remark 6.4), the node coordinates computed from the Bloch solution are real after a cyclic shift and satisfy (7); the residue ν_0 is then determined by the potential, through $P_n(0) = (-1)^m \omega^{-\nu_0}$. We do not prove the converse of Corollary 6.3 in general. The difficulty in proving existence for all odd q is that (7) is not settled by a global degree count: on the compactification $(\mathbb{RP}^1)^{2m}$ the E_ν are multilinear, hence sections of one and the same line bundle, and the corresponding Euler number vanishes. In the projective extension of Theorem 6.2 the 2^{2m} points with all $\varkappa_l \in \{0, \infty\}$ all give $V = 0$; the point with $\varkappa_l = \infty$ exactly for $l \in I$ is a zero of the system with $\nu_0 = s(I)$, and unless I is a union of pairs $\{l, q-l\}$ it is a degenerate zero whose local index we cannot compute in general.

7. Computations

All computations use the normal form of Lemma 2.1. For a lattice with dual group H (encoded by exponent vectors modulo the exponent m of H) we form the $N \times N$ matrices $\zeta_1 \cdots \zeta_d\,(xI - M_V(\zeta))$ and $\zeta_1 \cdots \zeta_d\,(xI - M_0(\zeta))$ over $\mathbb{Q}(e^{2\pi i/m})[x, \zeta, \widehat{V}]$, take the coefficients of the difference of determinants with respect to x and ζ, and compute a standard basis of the resulting ideal in the variables $\widehat{V}(h)$ with Singular. Rigidity is certified by checking that every $\widehat{V}(h)$ has a power in the ideal.

7.1. **Planar lattices.** Every planar lattice is $\Gamma = (a, 0)\mathbb{Z} \oplus (b, c)\mathbb{Z}$ with $0 \leq b < a$ and $ac = N$, and $g_1 = \gcd(a, b)$, $g_2 = c$. By Proposition 5.1, $\max(g_1, g_2) \geq 4$ implies nonrigidity.

Proposition 7.1. *Each of the* 64 *lattices with $2 \leq ac \leq 10$ and $\max(\gcd(a, b), c) \leq 3$ is rigid.*

The numbers of such lattices for $N = 2, \dots, 10$ are $3, 4, 5, 4, 10, 6, 9, 11, 12$; the computations are exact over the cyclotomic field and took at most 80 seconds each. This extends the range $ac \le 8$ reported in [4, Section 6]. Together with Proposition 5.1, Conjecture 1.2 holds for all planar lattices of index at most 10. For $N = 11$ the exact computation exceeded our time limit.

As evidence beyond this range we repeated the computation over a prime field $\mathbb{F}_p$, $p = 1000033$, in which the required roots of unity exist, restricting the unknowns to H° (legitimate by Corollary 3.3, since the one-dimensional projections have periods $g_1, g_2 \le 3$). All 26 lattices with $N \in \{11, 12\}$ and $\max(\gcd(a, b), c) \le 3$ are rigid modulo p. A modular computation is not a certificate, since a nonzero solution in characteristic 0 could reduce to the origin; we report it only as evidence for Conjecture 1.2.

7.2. **Cyclic lattices in $\mathbb{Z}^3$.** Let $H \subset \mathbb{T}^3$ be cyclic of order $N \le 7$. If no coordinate of a generator has order N, then (for $N \le 7$) all directional periods are at most 3 and Corollary 5.3 applies. If some coordinate of H is trivial, then $\Gamma = \Gamma' \oplus \mathbb{Z}$ up to order and Proposition 7.1 applies to Γ'. Otherwise, after permuting coordinates, $H = \langle (u, u^a, u^b) \rangle$ with $u = e^{2\pi i/N}$ and $1 \le a, b < N$. Up to permutations and inversions of coordinates there are 15 such classes with $4 \le N \le 7$ and $\max_j g_j \le 3$, and all of them are rigid; their directional periods range up to $(7, 7, 7)$. Hence every lattice in $\mathbb{Z}^3$ with cyclic dual group of order at most 7 and all axis indices at most 3 is rigid.

7.3. **Separability for $4\mathbb{Z} \oplus 2\mathbb{Z}$.** For $\Gamma = 4\mathbb{Z} \oplus 2\mathbb{Z}$ the full isospectrality ideal (seven Fourier variables) is zero-dimensional of vector-space dimension 201, and each of the three off-axis variables $\widehat{V}(h)$, $h \in H^\circ$, is nilpotent modulo the ideal. Hence every isospectral potential has the form $v(n_1)$ with v a 4-periodic isospectral potential. This is the first nontrivial instance of Conjecture 8.1 below.

7.4. **Conjugate-reflection symmetric potentials.** Conjecture 2 of [4] asserts that for every $q > 3$ there is a nonzero $q\mathbb{Z}$-periodic potential of the form $(v_1, v_2, \dots, \overline{v_2}, \overline{v_1})$ Floquet isospectral to 0. It was proved there for even q by symmetry operations on an explicit example of [5]. In one dimension the Floquet determinant equals $D_v(x) - z - z^{-1}$ with z-independent D_v, so isospectrality is equivalent to $\det(xI - C_q - \operatorname{diag} v) = \det(xI - C_q)$, where C_q is the adjacency matrix of the q-cycle, and also to the vanishing of the q coefficients of $D_v - D_0$. For odd $q = 2m + 1$ and $v = (x_1 + iy_1, \dots, x_m + iy_m, x_c, x_m - iy_m, \dots, x_1 - iy_1)$ with real unknowns, these coefficients are real, because D_v is invariant under reversal of v and $D_{\overline{v}} = \overline{D_v}$; this gives a square real system $F_q(x, y, x_c) = 0$ of q equations in q unknowns.

Proposition 7.2. *For $q = 5$, write $v = (x_1 + iy_1,\ x_2 + iy_2,\ x_3,\ x_2 - iy_2,\ x_1 - iy_1)$ with real unknowns. The five real polynomial equations expressing isospectrality have a nonzero real solution $x^* = (x_1, x_2, y_1, y_2, x_3)$ with*

$$x_1 = 1.364856976\dots, \quad x_2 = -0.787806159\dots, \quad x_3 = -1.154101634\dots,$$
$$y_1 = -1.162570196\dots, \quad y_2 = 1.340850081\dots$$

Consequently Conjecture 2 of [4] holds for every $q \in 5\mathbb{Z}$.

Proof. Let $F\colon \mathbb{R}^5 \to \mathbb{R}^5$ be the system and J its Jacobian matrix. Starting from the MSOLVE isolating box we computed a 60-digit approximation $\tilde{x}$ by Newton's method and set $X = \tilde{x} + [-10^{-20}, 10^{-20}]^5$. With $Y \approx J(\tilde{x})^{-1}$, the Krawczyk image $\tilde{x} - YF(\tilde{x}) + (I - YJ(X))(X - \tilde{x})$, evaluated in 60-digit interval arithmetic, lies in the interior of X in every coordinate (widths below 10^{-38}), so X contains a unique zero x^* of F; it is nonzero. Independently, MSOLVE reports exactly four nonzero real solutions, permuted by $v \mapsto -v$ and $v \mapsto \overline{v}$; we do not use this count. A lexicographic Gröbner basis gives the eliminant $x_3^{10}\, f(x_3^4)$ with $f(t) = t^7 + 30t^6 + 385t^5 + 2760t^4 + 11880t^3 + 22896t^2 - 6480t - 162000$, consistent with $f(x_3^4) \approx 0$ at $\tilde{x}$. For the consequence, if v is

5-periodic and isospectral, its $5r$-periodic repetition is isospectral by Lemma 2.3, and it has the conjugate-reflection form because $v_{6-i} = \overline{v_i}$ implies the same identity for the repetition. □

For larger q the coefficient system F_q is badly conditioned, and we use an equivalent formulation. Since D_v and D_0 are monic of degree q, $D_v - D_0$ has degree at most $q-1$, and $D_0(x_j) = 2\cos(\pi(2j+1)/2) = 0$ at the Chebyshev nodes $x_j = 2\cos(\pi(2j+1)/2q)$, $0 \le j \le q-1$. Hence $F_q = 0$ if and only if $G_q := (D_v(x_j))_{0\le j<q} = 0$, and the entries of G_q are real. We evaluate G_q and its Jacobian matrix from the transfer matrices at the nodes, using $\partial D_v(x)/\partial v_n = -(T_{n-1}\cdots T_1\, T_q \cdots T_{n+1})_{11}(x)$: in floating point for a damped Newton search from Gaussian random starts, and in midpoint-radius ball arithmetic (Arb [1], through python-flint), with the nodes enclosed in balls and with $6q + 400$ bits, for the refinement and for the Krawczyk test on the box of radius $2^{-3q-150}$ about the refined point.

Proposition 7.3. *For every prime q with $5 \le q \le 199$ and for $q = 9$ the system $G_q = 0$ has a nonzero real solution, certified by the Krawczyk test (Table 1). Consequently Conjecture 2 of [4] holds for every $q > 3$ that is even, or divisible by 9, or divisible by a prime p with $5 \le p \le 199$; in particular it holds for every $3 < q \le 210$. The smallest open period is $q = 211$.*

Proof. For each listed q the Krawczyk image lies in the interior of the box, which therefore contains a unique zero of G_q, hence of F_q; the zero is nonzero because the box is far from the origin (Table 1). Repetition preserves both isospectrality (Lemma 2.3) and the conjugate-reflection form, as in the proof of Proposition 7.2. An odd q with $3 < q \le 210$ either has a prime factor p with $5 \le p \le 199$ or is a power of 3 larger than 3, hence divisible by 9. Even q are covered by [4]. □

As an independent check, a separate script (`verify_point.py` in the supplementary archive) reads only the stored 30-digit points, refines each of them by Newton's method in ball arithmetic and repeats the Krawczyk test. It reproduces every certificate of Table 1, and in each case the certified zero differs from the stored point by less than $5 \cdot 10^{-30}$ in every coordinate. At the stored points themselves the relative defect $|D_v(x) - D_0(x)|/(1 + |D_0(x)|)$ at the $2q$ points $x = 2\cos(\pi(2k+1)/4q)$, which are not nodes, is below 10^{-20}; the loss against 10^{-30} reflects the size of the transfer-matrix products.

q	$\|u\|$	width	q	$\|u\|$	width	q	$\|u\|$	width
5	2.639	$4.80 \cdot 10^{-98}$	59	4.913	$5.80 \cdot 10^{-172}$	131	8.097	$5.41 \cdot 10^{-269}$
7	1.531	$2.04 \cdot 10^{-100}$	61	5.396	$1.37 \cdot 10^{-175}$	137	7.744	$6.82 \cdot 10^{-278}$
9	1.615	$1.59 \cdot 10^{-102}$	67	5.606	$1.41 \cdot 10^{-183}$	139	8.352	$1.94 \cdot 10^{-281}$
11	0.911	$4.03 \cdot 10^{-105}$	71	5.564	$8.62 \cdot 10^{-189}$	149	6.897	$1.18 \cdot 10^{-296}$
13	1.524	$1.10 \cdot 10^{-108}$	73	5.567	$1.52 \cdot 10^{-191}$	151	7.384	$8.49 \cdot 10^{-299}$
17	1.810	$1.11 \cdot 10^{-114}$	79	5.540	$1.51 \cdot 10^{-199}$	157	8.301	$4.99 \cdot 10^{-306}$
19	0.722	$1.13 \cdot 10^{-116}$	83	5.861	$7.80 \cdot 10^{-205}$	163	8.726	$9.22 \cdot 10^{-315}$
23	2.736	$3.34 \cdot 10^{-122}$	89	7.328	$6.07 \cdot 10^{-213}$	167	7.526	$1.05 \cdot 10^{-320}$
29	2.719	$1.66 \cdot 10^{-131}$	97	5.401	$8.31 \cdot 10^{-225}$	173	8.303	$1.24 \cdot 10^{-328}$
31	4.093	$9.37 \cdot 10^{-134}$	101	6.967	$9.35 \cdot 10^{-229}$	179	6.878	$5.83 \cdot 10^{-338}$
37	3.554	$2.82 \cdot 10^{-142}$	103	6.188	$6.79 \cdot 10^{-233}$	181	8.609	$1.18 \cdot 10^{-339}$
41	4.059	$9.70 \cdot 10^{-148}$	107	8.002	$1.48 \cdot 10^{-236}$	191	9.028	$3.64 \cdot 10^{-352}$
43	3.918	$4.05 \cdot 10^{-151}$	109	6.006	$2.92 \cdot 10^{-241}$	193	9.109	$3.94 \cdot 10^{-355}$
47	3.781	$6.64 \cdot 10^{-155}$	113	8.082	$5.39 \cdot 10^{-246}$	197	7.528	$1.59 \cdot 10^{-361}$
53	4.878	$1.39 \cdot 10^{-163}$	127	6.882	$1.23 \cdot 10^{-265}$	199	8.805	$1.90 \cdot 10^{-365}$

TABLE 1. Krawczyk certificates for Proposition 7.3. For each q: the norm of the certified point $u = (x, y, x_c)$ and the largest width of a coordinate of the Krawczyk image, which lies in the interior of the box of radius $2^{-3q-150}$ about the refined point (ball arithmetic with $6q + 400$ bits). The points, to 30 digits, are in the supplementary archive.

8. What remains open

(1) *Problem 1.1 for nonrectangular lattices.* By Theorem 5.9 it is equivalent to the off-axis vanishing statements (OV_d). The critical lattices are those to which neither Corollary 5.3 nor Theorem 5.5 applies. Among them are the cyclic planar lattices $\{n : n_1 + an_2 \equiv 0 \bmod N\}$ with $\gcd(a, N) = 1$ and $a \not\equiv \pm 1$, in which balanced elements are rare (there are none when $\gcd(a^2 - 1, N) = 1$) and only higher-order identities such as Proposition 5.11 are available; others have an axis index 2 or 3, for instance $(3, 0)\mathbb{Z} \oplus (1, 3)\mathbb{Z}$ and $(4, 0)\mathbb{Z} \oplus (1, 2)\mathbb{Z}$.
(2) *Splitting off* $3\mathbb{Z}$. Is $\Gamma' \oplus 3\mathbb{Z}$ rigid whenever Γ' is rigid? This holds for rectangular Γ' (Theorem 4.2) and for the diagonal lattices (Example 5.7), and fails for no known example.
(3) *Separation.* The complex analogue of Liu's real separability theorem [11] would give a complete description of the isospectral set.

Conjecture 8.1 (Separation)**.** *If V is Γ-periodic and Floquet isospectral to 0, then $V(n) = \sum_{j=1}^{d} v_j(n_j)$, where each v_j is $g_j(\Gamma)$-periodic and Floquet isospectral to 0 in one dimension.*

Conjecture 8.1 implies Conjecture 1.2: it suffices to prove it for rectangular lattices, since every Γ contains $m_1\mathbb{Z} \oplus \cdots \oplus m_d\mathbb{Z}$ (Lemma 2.3), and a separable Γ-periodic potential has $\operatorname{supp} \widehat{V} \subset \bigcup_j A_j$.

(4) *Conjecture 2 of* [4] *for the remaining odd periods*, that is, for odd $q > 3$ that are not divisible by 9 and have no prime factor p with $5 \le p \le 199$; the smallest is $q = 211$, and $q = 3 \cdot 211$ is another example; these periods have density $\frac{8}{9} \prod_{5 \le p \le 199} (1 - \frac{1}{p}) \approx 0.277$ among the odd integers. Every such q has a prime factor larger than 199 and repetition preserves the conjugate-reflection form, so by Corollary 6.3 it suffices to find a nonzero real solution of the multilinear system (7) for every prime $q \ge 211$. The numerical evidence is strong: the real solutions found by Newton's method from random starts, closed under the symmetries of Remark 6.4, number 4 (exactly, by msolve) and at least 120, 1024, 992 and 432 for $q = 5, 7, 9, 11, 13$, without visible sparse structure, and for every prime q with $5 \le q \le 199$ the search of Proposition 7.3 found a solution. A uniform existence argument is still missing; see Remark 6.5.

Data and code availability

The Singular and msolve inputs, the scripts that generate them, the interval and ball-arithmetic verification scripts (including the script used for Proposition 7.3, which records the certified points to 30 digits, and the independent re-verification script), the node-coordinate checks of Remark 6.5, and all logs accompany this manuscript as a supplementary archive.

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